

Particulate Matter Monitoring & Health Effects

Issue 2026-09-15 | PubMed entry window 15 September 2026 | 22 records screened in scope

22RECORDS IN SCOPE
(19 EXCLUDED, 4 HELD)**9**OF 10 SUBTOPIC
BANDS POPULATED**9**EFFECT ESTIMATES
(ALL WITH A CI)**8**RECORDS CARRYING
NO PMID**5**TIER-A
RECORDS

Signal of the day

Per unit mass, wildfire $\text{PM}_{2.5}$ carries about 4.6 times the mortality hazard of ordinary urban $\text{PM}_{2.5}$ — but the exposure contrast behind that number is a thirtyfold difference in concentration, and that is the part to read carefully.

Su et al. (*Lancet Oncol*) followed **414,016** SEER-Medicare patients aged 65 and over with histologically confirmed lung cancer through **1,211,490** person-years (2008–2019), with zip-code wildfire and non-wildfire $\text{PM}_{2.5}$ from validated models and a time-varying Cox model on 3-year moving averages. Per $1 \mu\text{g}/\text{m}^3$, the hazard of death was **HR 1.0782 (95% CI 1.0641–1.0924)** for wildfire $\text{PM}_{2.5}$ against **1.0171 (1.0149–1.0192)** for non-wildfire — non-overlapping intervals, and the sharpest population-scale statement yet that smoke mass and urban mass are not the same exposure.

The caution is arithmetic. Median wildfire $\text{PM}_{2.5}$ was **$0.28 \mu\text{g}/\text{m}^3$** (IQR 0.21–0.39) against **8.11** for non-wildfire, so a “per $1 \mu\text{g}/\text{m}^3$ ” wildfire coefficient is extrapolated several IQR-widths beyond anything observed, and on the per-SD scale the ordering reverses (1.0229 versus 1.0384) because an SD of wildfire $\text{PM}_{2.5}$ is a far smaller absolute dose. Both numbers are correct; only one is a statement about the air people breathed. The monitoring implication survives either reading: a network reporting mass alone cannot distinguish these two exposures.

What changed today

- ▶ **A dense day, and for once the epidemiology carries intervals.** Twenty-two records passed screening against sixteen on 14 September, and **nine** carry a published exposure–response estimate with a 95% interval. Eight of the nine are PM estimates proper — the first issue in some weeks whose forest plot is not carried by heat or co-pollutant surrogates. Nine of ten subtopic bands are populated; *Burden, policy & mitigation* is the sole empty one.
- ▶ **The DOI gate removed three duplicates that title screening called new, and one is a repeat offender.** The Consensus sweep returned ten low-cost-sensor calibration hits; four matched corpus titles directly, and the DOI check then removed three more — a multi-city PurpleAir/OPC-N3 calibration, the Sofia calibration-transfer study, and the Bogotá T640X node calibration. That last record was *already removed on 14 September* for the same reason: it shipped on 3 August under a completely rewritten title. Consensus keeps re-surfacing it and no title-normalised index will ever catch it.
- ▶ **Two findings rest on weak arithmetic, and both are named in place.** The Himawari-8 study of Javanese coal-plant plumes reports random forest and XGBoost scoring **exactly 1.000** on every metric — but the target is a deterministic threshold on the two input variables, so the models relearn the labelling rule and the 2026–2060 projection built on them is uninterpretable. The Korean agricultural analysis lifts precursor-PM_{2.5} correlations to $r \geq 0.816$ by dividing every series by one wind-speed-derived exchange rate, which is what dividing several series by a common quantity does. A third belongs with them: the *JECH* suicide meta-analysis pools per-increment and high-versus-low categorical PM_{2.5} contrasts under one label, producing a 38-fold short-versus-long gap that is not a duration effect.
- ▶ **Four in-scope records were verified and deliberately held.** The MoAE low-cost sensor calibration framework (*Atmos Environ*), a CTM–InMAP burden study over Punjab, a SW-Europe dust and sea-salt compliance analysis (*Atmos Res*) and an office breathing-zone sensor-placement chamber study (*Build Environ*) all have resolving DOIs and no abstract in Crossref, Europe PMC, Semantic Scholar, Consensus or Scholar Gateway — all four same-day Elsevier deposits, the failure mode logged on 14 September. They are kept *out of seen.json*; the MoAE paper is the tier-A pickup. Separately, **arXiv returned HTTP 406** and an empty body, a new failure distinct from the `submittedDate` behaviour previously logged, and OpenAlex returned HTTP 429 again.
- ▶ **Evidence geography is tied, not led.** China and Global / multi-region sit at **seven records each**. Sub-Saharan Africa’s two matter disproportionately — a Nigerian cohort and South African pneumococcal surveillance, both built on exposure surfaces with no in-country reference network.

Corpus analytics

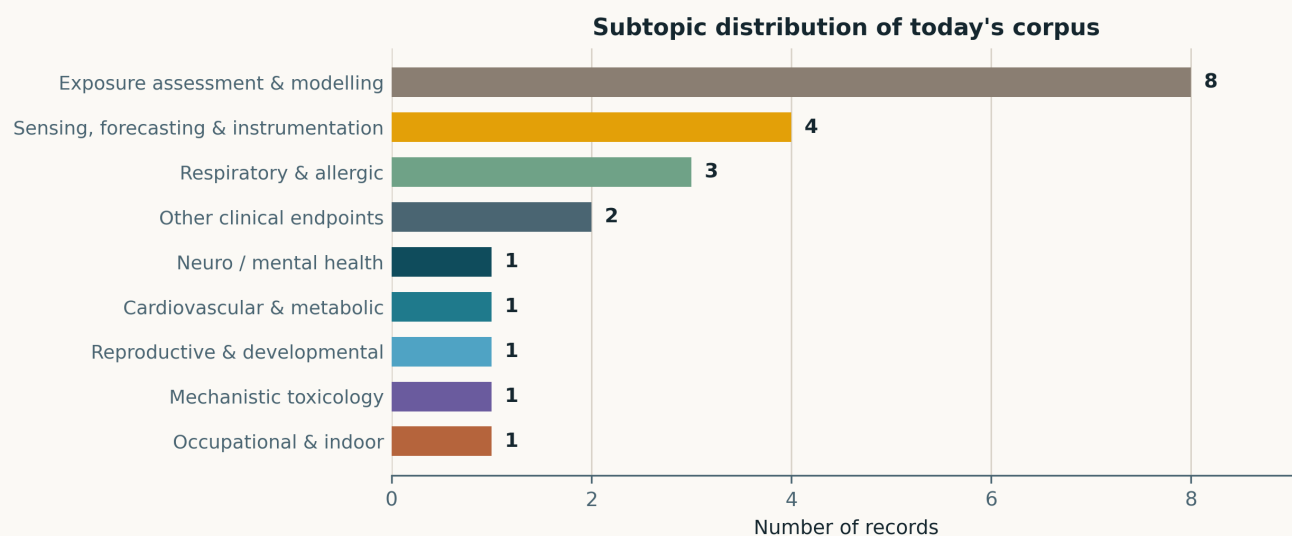


Fig. Subtopic assignment of the 22 in-scope records, single-label by dominant contribution. *Exposure assessment & modelling* takes 8, twice the next cluster; *Sensing, forecasting & instrumentation* 4; *Respiratory & allergic* 3; *Other clinical endpoints* 2; and *Neuro / mental health*, *Cardiovascular & metabolic*, *Reproductive & developmental*, *Mechanistic toxicology* and *Occupational & indoor* one each. Nine of the ten standing bands are populated — *Burden, policy & mitigation* is the sole empty one.

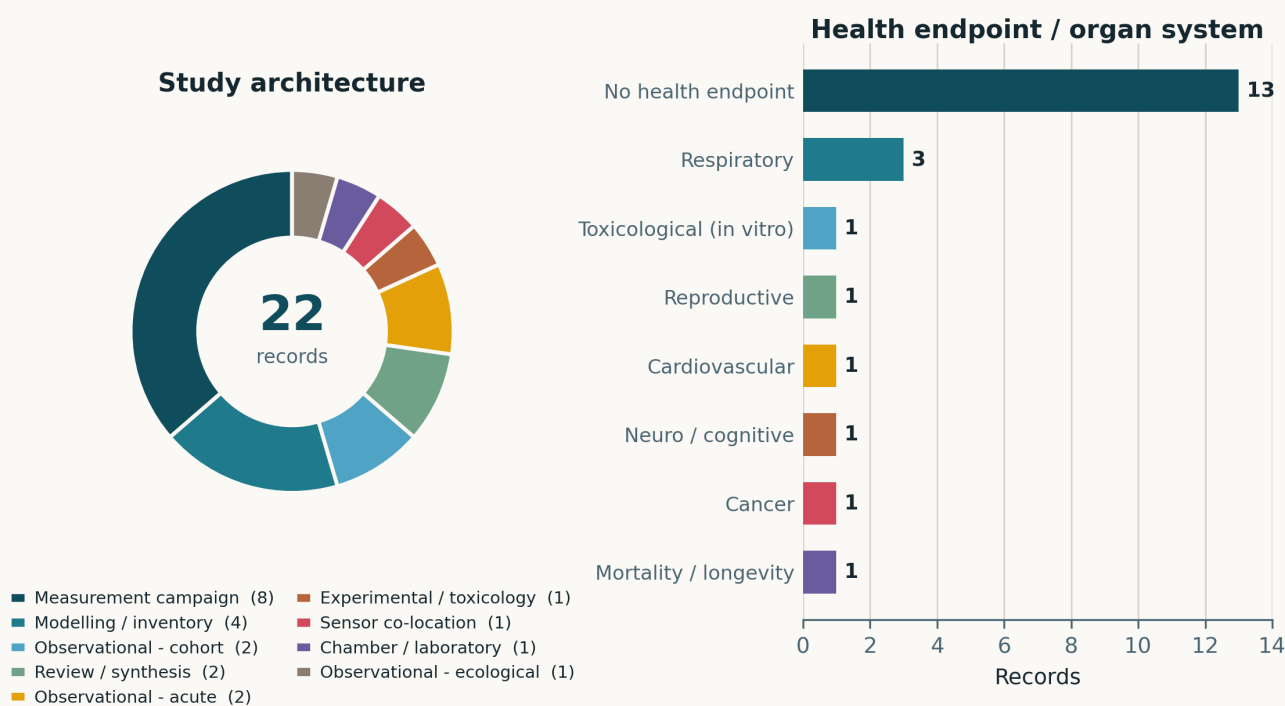


Fig. Left — study architecture. Measurement campaign is the largest class at 8, ahead of modelling and inventory at 4; observational designs split 2 cohort, 2 acute and 1 ecological, with 2 reviews and one each of experimental toxicology, sensor co-location and chamber work. Right — health endpoint. **Thirteen of the 22 records carry no health endpoint**, which is the aerosol-chemistry, instrument and modelling half of the corpus; among those that do, Respiratory leads with 3 and the remaining six endpoint classes — toxicological, reproductive, cardiovascular, neuro/cognitive, cancer and mortality — hold one record each.

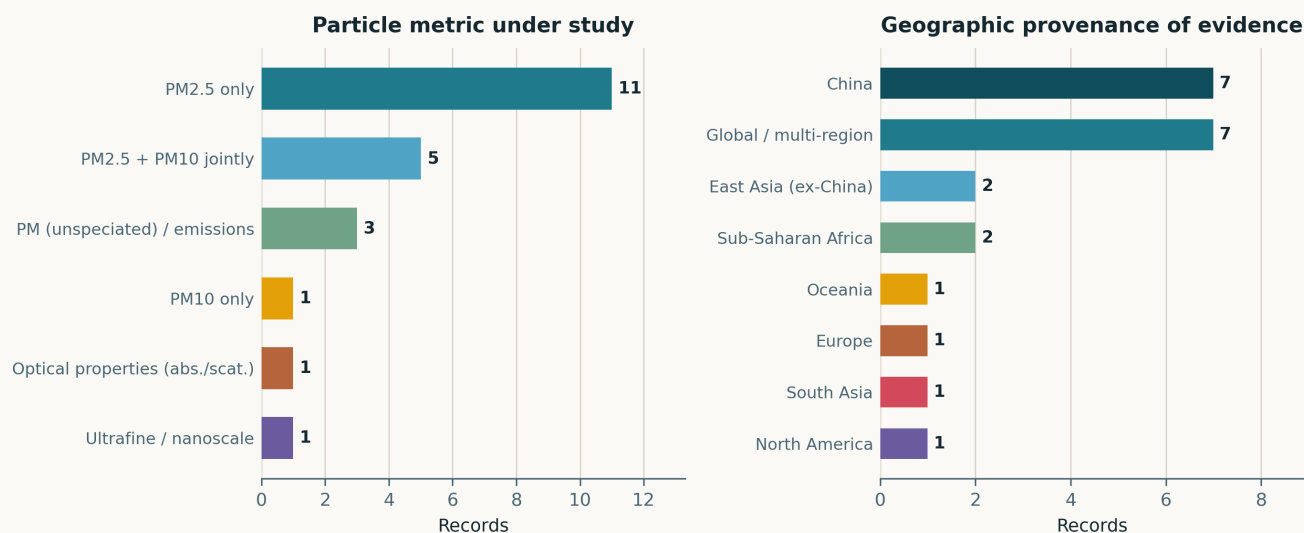


Fig. Left — particle metric. PM_{2.5} alone is the exposure in 11 of 22 records and PM_{2.5} with PM₁₀ jointly in a further 5; ultrafine, optical-property and PM₁₀-only work contribute one record each, so size-resolved and optical measurement remain thin against mass. Right — geographic provenance. China and Global / multi-region are tied at seven records each; no single country dominates this issue.

Life-course windows addressed today (records may span several stages)

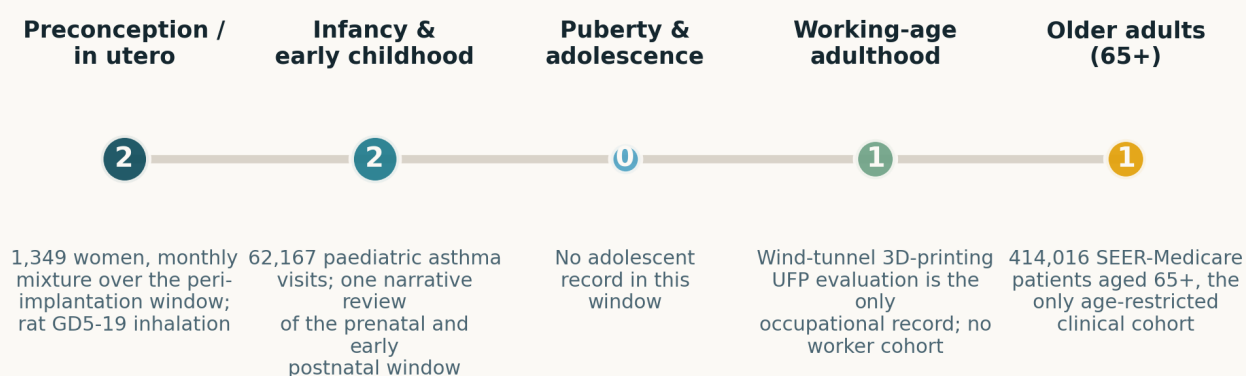


Fig. Life-course windows addressed. Four of the five stages are occupied and **only puberty and adolescence is empty**, an improvement on the three-empty issues of the past fortnight. Working-age adulthood is nominally filled but by a wind-tunnel instrument evaluation rather than any worker cohort, so the occupational gap is real even where the marker is not zero.

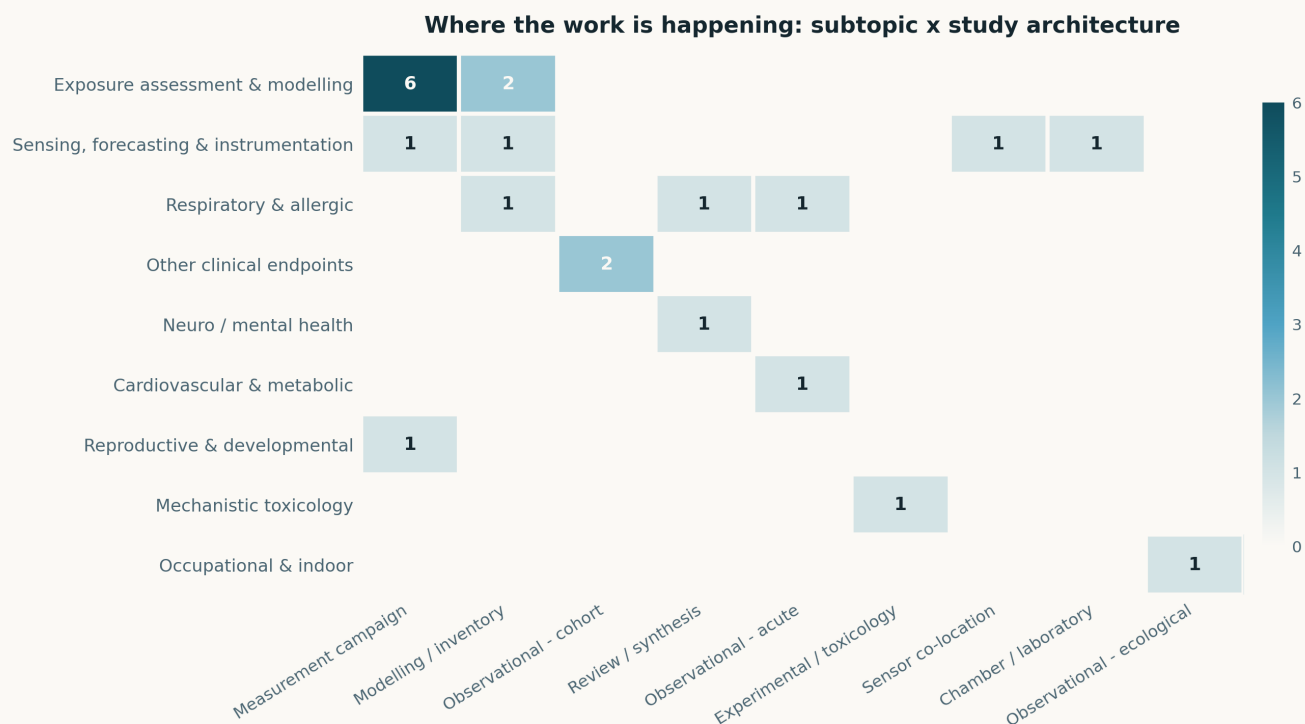


Fig. Subtopic × study-architecture cross-tabulation. **Fifteen of the 81 cells are occupied.** Density sits where measurement campaigns meet exposure assessment; each clinical row is a single cell, which is what a one-record-per-endpoint day looks like.

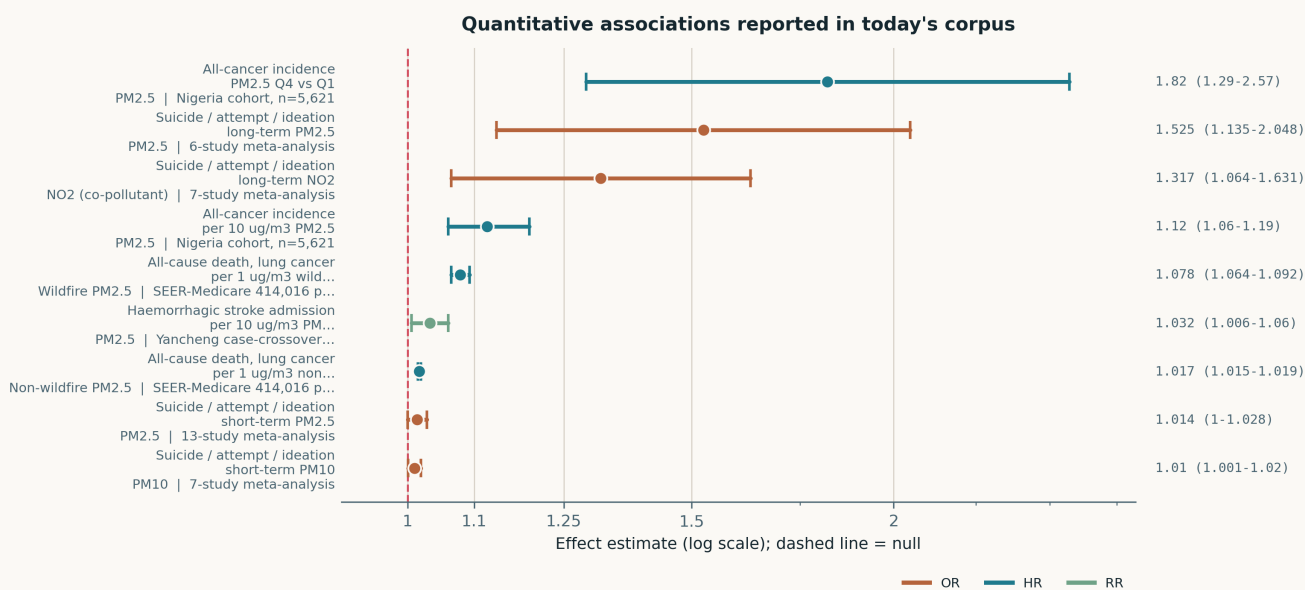


Fig. All nine interval-bearing estimates in today's corpus on a common logarithmic axis. **Eight are PM exposure-response estimates** — two wildfire versus non-wildfire mortality hazards, two all-cancer incidence hazards, one haemorrhagic-stroke risk ratio and three suicide-outcome odds ratios for PM_{2.5} and PM₁₀ — with the NO₂ estimate shown as a co-pollutant comparator. Interval width spans a factor of about 300, from the SEER-Medicare non-wildfire estimate (1.0149–1.0192) to the Nigerian top-quartile hazard (1.29–2.57); that is a difference in study size and exposure contrast, not in precision of measurement. The two short-term suicide estimates at the foot of the plot and the long-term estimate in second place come from the same meta-analysis on incompatible exposure scales and should not be read against each other.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Exposure assessment & modelling

8 records

Han et al. 2026 Environ Pollut

Source apportionment of PM_{2.5} oxidative potential in metropolitan Melbourne

Published PMF source contributions linked by multiple linear regression to oxidative potential measured on **three complementary acellular assays** (DTT, AA, DCFH); models explain **73–86%** of OP variance. Combustion dominates OP_v-DTT at 52–53% with biomass burning above traffic; OP_v-AA is driven by metal-rich sources including diesel at 20–22%; secondary nitrate contributes 28–33% and secondary sulphate 10% at the second site. The assay-dependent ranking is the result — a single-assay OP metric would nominate a different mitigation target. Reusing an earlier PMF solution propagates its rotational ambiguity, and no uncertainty is quoted on the source-resolved shares. First Southern Hemisphere estimate of its kind.

doi: 10.1016/j.envpol.2026.129159

Liu et al. 2026 J Environ Manage

Decadal changes in PM_{2.5} sources and secondary aerosol drivers, Shenzhen 2014–2024

Eleven years of speciated filter sampling with PMF apportionment and machine-learning attribution of secondary formation drivers. Annual mean fell **37.7 to 17.3 $\mu\text{g m}^{-3}$** . Secondary sulphate was historically dominant and declined fastest; vehicles became the largest single contributor after 2019; **secondary nitrate barely moved and is now the binding constraint**. NO_x and VOC cuts drove nitrate and SOA declines but rising atmospheric oxidation capacity offset them, while sulphate tracked oxidation and regional transport over local SO₂. Attributing a decadal trend to oxidation capacity from a correlational model at one site cannot exclude meteorological confounding, and the PMF factor series carry no uncertainty bounds.

doi: 10.1016/j.jenvman.2026.130913

Zhang et al. 2026 Environ Pollut

Surface-based temperature inversion climatology across China and its relation to PM_{2.5}

Hourly microwave-radiometer profiles from **33 stations** over 2024–2025, validated against radiosondes at 25 sites, used to build a surface-based inversion climatology and relate it to PM_{2.5} at hotspot stations. Winter SBI occurrence **39.2%** against **12.1%** in summer, with a northwest-to-southeast gradient peaking in the Tarim Basin and Hexi Corridor. SHAP and Spearman analysis place boundary-layer variables in the top predictor tier monthly and daily, with effective mixing-layer height third among all hourly predictors in every season. Two years is short for a climatology, the network is sparse relative to the terrain it spans, and SHAP importance over correlated meteorological predictors does not establish the causal pathway it is read as establishing.

doi: 10.1016/j.envpol.2026.129172

Kim et al. 2026 PLoS One

LASSO-based reduced-form CMAQ for ozone and PM_{2.5} response to emission change

Surrogate for CMAQ fitted by LASSO over **118 emission scenarios** spanning seven sectors and 17 provincial regions, with an adaptive logit transform to respect concentration bounds. Trained on 100 simulations and tested on 18: mean RMSE **0.1 ppb** for ozone and **0.1 $\mu\text{g m}^{-3}$** for PM_{2.5} against the parent model. The sparse set of retained sector–region terms is the practical output, making the control-design problem interpretable rather than opaque. Error is quoted against CMAQ and not against observations, so every CMAQ bias passes through undiagnosed; 118 scenarios is thin for a 119-dimensional predictor space and no out-of-envelope extrapolation test is reported.

doi: 10.1371/journal.pone.0347073

Aji et al. 2026 Environ Monit Assess

Fine-mode combustion aerosol events near Indonesian coal plants from AE-filtered Himawari-8

41,119 daily Himawari-8 observations around 13 Java coal plants, 2015–2025, with events defined as Ångström exponent > 1.2 and AOD > 0.4 (8.52% of days). Suralaya has the highest mean AOD at 1.01 with 51.4% very-high-pollution days; Pacitan the lowest at 0.46. Random forest and XGBoost return accuracy, precision, recall, F1 and AUC-ROC all **exactly 1.000** — on a target that is a deterministic threshold function of the two inputs, so the models relearn the labelling rule rather than predict anything, and the 2026–2060 Monte Carlo projection built on them is uninterpretable. Column AOD without surface validation cannot separate plant plume from boundary-layer aerosol.

doi: 10.1007/s10661-026-15804-1

Ilies et al. 2026 Front Public Health

Urban PM₁₀ and population exposure from satellite, meteorological and socio-spatial data

MODIS-MAIAC AOD combined with four ground stations and meteorology in a regression for PM₁₀ over Oradea, Romania, then folded with population density, built form, NDVI and elevation into a composite exposure index. Cold-season maxima reach **41.8 $\mu\text{g m}^{-3}$** against an annual mean near 20.5 — above WHO, inside EU limits — and **20.1%** of the metropolitan population falls in high-predisposition areas. The AOD–PM₁₀ relation alone is weak (**R = 0.28**) and meteorology carries the model, which is the authors' own admission and the standing reason satellite-only surfaces underperform at city scale. Four stations cannot validate an intra-urban field, and the composite index is unweighted and never checked against measured personal exposure.

doi: 10.3389/fpubh.2026.1940025

Kim & Park 2026 Atmosphere

Loading-rate-based analysis of PM_{2.5} and precursor gases in Korean agricultural areas

547 days of hourly data from eight agricultural sites, comparing concentration-based with loading-rate-based interpretation, where loading rate is concentration scaled by a wind-speed-derived air exchange rate. Concentration correlations among PM_{2.5}, NH₃, NO₂ and SO₂ are weak and inconsistent; loading-rate correlations are strong and uniform at $r \geq 0.816$, and the diurnal shape flips from bimodal to unimodal. The authors concede the improvement may reflect the shared wind-speed term — it almost certainly does, since dividing every series by one common quantity induces correlation by construction. Recorded as a method to treat with suspicion: any dense open-field network normalising by a common meteorological variable manufactures the same result.

doi: 10.3390/atmos17090898

Ahmad et al. 2026 Atmosphere*Carbonaceous composition and multi-wavelength optical properties of PM in Lahore*

Six-week urban campaign in Lahore, 30 quartz filters collected, 25 usable for gravimetry and optics, **only the first 15 analysed for OC/EC**. $\text{PM}_{2.5}$ ranged **40–417 $\mu\text{g m}^{-3}$** with a mean of **137 $\pm$ 84**. Organic carbon was 76.6% of total carbon against 23.4% elemental. Mean absorption Ångström exponent **1.17 $\pm$ 0.30**, with 56% of samples in 1.0–1.5, 32% below 1.0 and 12% above 1.5. The authors are appropriately careful that AAE intervals describe spectral behaviour rather than attribute sources. Fifteen OC/EC filters in one pre-monsoon season cannot characterise Lahore's annual cycle, and the gravimetric series has no co-located reference for traceability. Directly relevant to optical low-cost sensors, whose response depends on exactly this absorption behaviour.

doi: [10.3390/atmos17090899](https://doi.org/10.3390/atmos17090899)

Sensing, forecasting & instrumentation

4 records

Vance et al. 2026 J Occup Environ Hyg*Evaluation of the Partector 2 Pro for ultrafine particles from 3D printing*

Wind-tunnel comparison of a portable diffusion-charging monitor against SMPS and CPC during controlled printing of PLA, PETG, TPU and nylon. Number-concentration agreement is strong across all four filaments — regression slopes **0.88–1.05**, Pearson $r > 0.96$, mean deviations **–9.6% to +6.2%**, inside the $\pm 30\%$ manufacturer specification. Sizing is systematically biased low (slopes 0.82–0.90) and worst at the fine end: **–27.0% to +7.5% at D10** and **–28.3% to –7.3% across D25–D75**. That pattern is the operational point for any distributed ultrafine network — the instrument is fit for concentration surveillance but its size distribution must not be used to infer deposition fraction. Wind tunnel only, with no humidity or co-pollutant interference testing.

doi: [10.1080/15459624.2026.2724020](https://doi.org/10.1080/15459624.2026.2724020)**Wathore et al. 2026** Aerosol Air Qual Res*Environmental and temporal performance of ML calibration for a low-cost $\text{PM}_{2.5}$ sensor*

Long-term co-location in Vishakhapatnam comparing Random Forest and XGBoost against linear regression, adding temperature, relative humidity, hour-of-day and month-of-year stepwise, plus a hybrid ensemble. The best single model cuts RMSE **58%** against the linear baseline and the hybrid reaches **63%**, meeting US EPA low-cost $\text{PM}_{2.5}$ performance criteria. The contribution is the evaluation design rather than the models: performance is reported binned across environmental and temporal strata and across AQI categories, which exposes where a calibration degrades instead of burying it in a pooled R^2 . No absolute RMSE in $\mu\text{g m}^{-3}$ is given, and single-site training leaves transferability untested — the same limitation already flagged for the Sofia calibration-transfer work in this corpus. *Connector-sourced; outside the 15 September entry window.*

doi: [10.1007/s44408-026-00094-6](https://doi.org/10.1007/s44408-026-00094-6)**Komiljon et al. 2026** Atmosphere*Non-cumulative, size-specific calibration of low-cost PM sensors under simulated drilling*

Controlled laboratory evaluation of PMS5003 and Sniffer4D under simulated concrete-slab drilling, with six correction models — linear, polynomial, Random Forest, XGBoost, ANN, Kalman — fitted independently per size fraction. Random Forest and ANN perform best, $R^2 > 0.89$ **for PM_1** and **> 0.87 for $\text{PM}_{2.5}$** at 50 s averaging. The premise is the useful part: reporting non-cumulative rather than cumulative bins, since cumulative output conflates fractions with different deposition behaviour and different regulatory meaning. Against that, a single drilling source in a chamber produces a mineral-dust size distribution that will not transfer to combustion aerosol, and PM_{10} performance is not reported alongside the finer fractions. *Connector-sourced; Crossref resolves only the preprint of record.*

doi: [10.20944/preprints202604.1260.v1](https://doi.org/10.20944/preprints202604.1260.v1)**Wang et al. 2026** Environ Sci Technol*14-day $\text{PM}_{2.5}$ and O_3 forecasting over eastern China with a CNN–Transformer*

Hybrid architecture combining multilevel regional meteorological fields with emission inventories at a 14-day lead. Relative R^2 gains over baseline are **40% for O_3** (0.58 to 0.81) and **18% for $\text{PM}_{2.5}$** (0.44 to 0.52); 850 hPa fields add 5% for O_3 and emissions add 9.8% for $\text{PM}_{2.5}$; 66–83% of O_3 episodes are captured in the north. Read honestly, $\text{PM}_{2.5}$ skill remains poor — R^2 0.52 at 14 days is not decision-grade, and the paper's own framing concedes O_3 is the stronger case. No episode-capture rate is reported for $\text{PM}_{2.5}$, which is the number a forecast user would act on, and gridded reanalysis truth hides station-scale error.

doi: [10.1021/acs.est.6c05923](https://doi.org/10.1021/acs.est.6c05923)

Respiratory & allergic

3 records

Kremer et al. 2026 Nat Microbiol*Pneumococcal population structure modifies the effect of air pollution on invasive disease*

Nineteen years of South African national surveillance — **59,017** invasive pneumococcal disease cases with **4,350** sequenced isolates — in Bayesian spatiotemporal models linking environment to pathogen diversity. PM_{2.5} is positively associated with IPD, but **the lag structure varies by serotype**: only serotypes 4, 8 and 23F confer immediate risk, and high prevalence of lineage GPSC21 (serotype 19F) shifts the population-level response to immediate. Cumulative 8-week effects for moderate humidity and cold minimum temperature are +5% and +4%. No confidence interval accompanies the headline pollution association and the serotype-stratified lag models present a large multiple-comparison surface. Still the first demonstration that pathogen genotype modifies an air-quality exposure–response, and a direct argument for pairing dense PM monitoring with pathogen surveillance.

doi: 10.1038/s41564-026-02458-5

Zhu et al. 2026 J Asthma Allergy*Extreme cold events and paediatric asthma visits, modified by ambient CO and SO₂*

62,167 paediatric asthma outpatient visits at one tertiary hospital in Nanjing, 2019–2024, quasi-Poisson DLNM over lag 0–14. Extreme cold events show **no overall association**, but cumulative RR reaches **2.60 (95% CI 1.13–5.95)** in the high-CO stratum and **3.41 (1.15–10.11)** in the high-SO₂ stratum. The intervals are very wide and the strata are subsets of an already null main effect, so this is hypothesis-generating; notably PM_{2.5} and PM₁₀ did not emerge as modifiers. Single-hospital catchment with city-average exposure. Relevant here mainly as a reminder that pollutant-by-meteorology interaction terms demand far denser exposure data than one municipal station provides.

doi: 10.2147/jaa.s483101

Martinez & Nadeau 2026 Curr Allergy Asthma Rep*Early-life exposure to air pollutants and childhood asthma*

Narrative synthesis of recent cohort and mechanistic work on the prenatal and early postnatal window. Reports epigenetic modification in airway epithelium, a convergence point shared between tobacco smoke exposure and genetic asthma predisposition, new serum biomarkers of exposure, and a murine result in which asthma-predisposing information passes from exposed dam to offspring **via NK cells**. No pooled estimate, no effect size, and no stated search strategy or inclusion criteria, so study selection is unauditably and the piece cannot be weighted against the systematic reviews in this window. Useful as a pointer to the NK-cell transmission mechanism and the biomarker candidates, and for nothing quantitative.

doi: 10.1007/s11882-026-01293-2

Other clinical endpoints

2 records

Su et al. 2026 Lancet Oncol*Wildfire-related PM_{2.5} and overall survival of older lung cancer patients*

414,016 SEER-Medicare patients aged 65+, **1,211,490** person-years, 2008–2019, time-varying Cox on 3-year moving averages of zip-code wildfire and non-wildfire PM_{2.5}. Per 1 µg m⁻³: **HR 1.0782 (95% CI 1.0641–1.0924)** for wildfire against **1.0171 (1.0149–1.0192)** for non-wildfire, a 4.6-fold difference in per-unit potency. The exposure contrast is the weakness — median wildfire PM_{2.5} was 0.28 µg m⁻³ (IQR 0.21–0.39) against 8.11 for non-wildfire, so the per-unit hazard is extrapolated far outside the observed range, and the per-SD estimates (1.0229 versus 1.0384) reverse the ordering. Zip-code assignment with no residential history. Species-resolved rather than mass-only monitoring is the direct implication.

doi: 10.1016/s1470-2045(26)00295-0

Mtenga et al. 2026 Cancer Epidemiol*Ambient PM_{2.5} exposure and cancer incidence in Nigerian adults*

First prospective sub-Saharan African cohort on this question. 5,621 adults, median follow-up 4.7 years, 147 incident cancers confirmed histopathologically, mean PM_{2.5} **50.4 µg m⁻³** — tenfold the WHO guideline. Each 10 µg m⁻³ gives **HR 1.12 (95% CI 1.06–1.19)** for all-cancer incidence and the top quartile **1.82 (1.29–2.57)**; mediation analysis attributes **41.3%** to oxidative stress and inflammation. Site-specific signals for lung, breast and haematopoietic cancer rest on very small strata at 147 total events, and biomarkers measured once in a subsample cannot establish the temporal ordering mediation requires. The 1 km satellite–land-use regression exposure surface has no ground reference network to anchor it.

doi: 10.1016/j.canep.2026.103232

Neuro / mental health

1 record

Pearson et al. 2026 J Epidemiol Community Health*Environmental exposure and risk of suicide, attempted suicide and ideation*

Forty-five studies screened across air, noise, light and water pollution, 29 pooled. Short-term PM_{2.5} OR 1.0139 (95% CI 1.0001–1.0279) over 13 studies and PM₁₀ 1.0100 (1.0007–1.0195) over 7; long-term PM_{2.5} 1.5254 (1.1351–2.0481) over 6 and NO₂ 1.3173 (1.0641–1.6308). The short-term PM_{2.5} lower bound sits at 1.0001, so significance is nominal. More seriously, the **38-fold gap between short- and long-term PM_{2.5} point estimates is an artefact of pooling per-increment against high-versus-low categorical contrasts under one label**; the two are not on a common scale and must not be read as a duration–response. Newcastle-Ottawa assessment only, with no weighting for exposure misclassification.

doi: 10.1136/jech-2025-225491

Cardiovascular & metabolic

1 record

Xu et al. 2026 Front Public Health*Ambient fine particulate matter and haemorrhagic stroke in Yancheng, China*

Individual-level time-stratified case-crossover on 2,312 haemorrhagic stroke admissions across four Yancheng hospitals, 2022–2024, conditional logistic regression. Strongest at lag01: each 10 µg m⁻³ of PM_{2.5} associated with a **3.24% increase (95% CI 0.55–6.00)**, with lag0 2.50% (0.11–4.95) and lag1 2.76% (0.32–5.27). The association survives adjustment for O₃ but **loses significance once SO₂ or NO₂ enters** — which is the finding that matters, because it means the PM_{2.5} coefficient is not separable from co-emitted combustion products at this station density. City-average exposure assigned to residence leaves within-city gradients unmeasured, and a four-hospital catchment is not the full case population.

doi: 10.3389/fpubh.2026.1882030

Reproductive & developmental

1 record

Sun et al. 2026 Environ Res*Air pollutant mixtures and endometrial receptivity in the peri-implantation window*

1,349 women across 14 cities in Liaoning, with quantile g-computation and BKMR on a six-pollutant mixture against sonographic endometrial thickness and volume. A simultaneous one-quartile increase in the mixture is associated with **0.5 mm** less thickness and **0.28 ml** less volume, with CO and O₃ carrying the largest weights and PM_{2.5} and PM₁₀ not dominating. **No confidence intervals are reported for the mixture effects**, which for a BKMR result is a substantive omission. A 0.5 mm shift is at or below the reproducibility of transvaginal endometrial measurement, so the clinical meaning is unclear even if the association holds. Monthly gridded exposure with no personal or indoor component, against an implantation window measured in days.

doi: 10.1016/j.envres.2026.125682

Mechanistic toxicology

1 record

Garner et al. 2026 Reprod Toxicol*Gestational nanoparticle inhalation disrupts placental zone structure in rats*

Pregnant Sprague Dawley rats exposed by whole-body inhalation to nano-TiO₂ at **9.23 ± 0.39 mg m⁻³** from gestational day 5 to 19, placentas harvested GD20 for histology. Placental weight and placental efficiency were **unchanged**, but decidua and labyrinth zone sizes fell, with compensatory increases in blood space number and maternal blood space area. The compensation is the interesting part: structural remodelling without a mass-level phenotype means placental weight is a poor screening endpoint for ultrafine exposure. The exposure is roughly three orders of magnitude above ambient ultrafine mass and engineered TiO₂ is a poor surrogate for combustion ultrafines in surface chemistry, so this bounds mechanism and not risk. No litter-level statistics are reported.

doi: 10.1016/j.reprotox.2026.109361

Occupational & indoor

1 record

Chen et al. 2026 Indoor Air*Indoor air protection under extreme air pollution risk: evidence from a sandstorm in China*

Difference-in-differences on Baidu search intensity for air purifiers and indoor air-quality information around China's most intense sandstorm in a decade (early 2021). Affected regions show roughly **18%** higher search intensity than unaffected ones, rising to about **33%** in historically cleaner areas and about **200%** where baseline purifier sales were lowest. The gradient is the useful result: protective response scales inversely with prior exposure experience and prior equipment stock. Search volume proxies intent, not purchase, use, or any realised reduction in indoor PM_{2.5} — no concentration data enter the study at all — and regional treatment assignment with no individual exposure makes the effect ecological throughout.

doi: 10.1155/ina/2274208

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
45 studies, 29 pooled, 2010-2024	Environmental exposure and risk of death by suicide, attempted suicide and suicide ideation: a systematic review and meta-analysis.	<i>J Epidemiol Community Health</i>	Systematic review (PRISMA)	Ambient PM _{2.5} and PM ₁₀ , short- and long-term contrasts pooled together	Global / multi-region
Cardiovascular & metabolic					
Yancheng, 2,312 admissions 2022-2024	Association between ambient fine particulate matter and hemorrhagic stroke in Yancheng, China.	<i>Front Public Health</i>	Time-stratified case-crossover	Station-average daily PM _{2.5} , lag 0 to lag 01	China
Reproductive & developmental					
Liaoning fertility clinics, n=1,349	Ambient air pollutant mixtures in relation to sonographic endometrial receptivity parameters during the peri-implantation window: a clinic-based observational study.	<i>Environ Res</i>	Observational exposure study	Monthly mean PM _{2.5} , PM ₁₀ , CO, NO ₂ , SO ₂ , O ₃ from a high-resolution reanalysis grid	China
Respiratory & allergic					
Nanjing, 62,167 paediatric asthma visits	Extreme Cold Events and Pediatric Asthma Outpatient Visits: The Modifying Role of Ambient Carbon Monoxide and Sulfur Dioxide.	<i>J Asthma Allergy</i>	Time-series	Daily PM _{2.5} and PM ₁₀ as stratifying co-pollutants, not the exposure of interest	China
Narrative review, prenatal and early postnatal window	Early Life Exposure to Air Pollutants and Childhood Asthma.	<i>Curr Allergy Asthma Rep</i>	Narrative review	Prenatal and early postnatal ambient air pollutants, reviewed qualitatively	Global / multi-region
South Africa, 59,017 IPD cases, 4,350 genomes	Pneumococcal population structure influences the effects of air pollution on invasive disease risk in South Africa.	<i>Nat Microbiol</i>	Bayesian spatiotemporal model	Ambient PM _{2.5} , lagged to 8 weeks, linked to serotype and lineage	Sub-Saharan Africa
Mechanistic toxicology					
Sprague Dawley whole-body inhalation, GD5-19	Gestational inhalation of nanoparticles disrupts placental zone structure and induces vascular placentation in rats.	<i>Reprod Toxicol</i>	Animal model	nano-TiO ₂ aerosol, 9.23 +/- 0.39 mg/m ³ whole-body, as an ultrafine PM proxy	Laboratory
Occupational & indoor					
China 2021 sandstorm, Baidu search DID	Indoor Air Protection Under Extreme Air Pollution Risk: Evidence From a Sandstorm in China.	<i>Indoor Air</i>	Ecological panel	Sandstorm-driven outdoor PM episode as an exogenous shock; no PM measured directly	China
Sensing, forecasting & instrumentation					
Central-eastern China, 14-day lead time	14-Day PM _{2.5} and O ₃ Forecasting over Eastern China Using a Hybrid CNN-Transformer Architecture Driven by Multilevel Meteorological Fields.	<i>Environ Sci Technol</i>	Deep-learning model	Gridded surface PM _{2.5} and O ₃ forecast from multilevel meteorology plus emissions	China
PMS5003 and Sniffer4D, concrete drilling	Non-Cumulative, Size-Specific Calibration of Low-Cost Particulate Matter Sensors Under Simulated Construction Drilling Events.	<i>Atmosphere (preprint of record)</i>	Chamber / laboratory	Non-cumulative PM ₁ , PM _{2.5} , PM ₁₀ fractions under simulated concrete-slab drilling	Not stated
Partector 2 Pro vs SMPS and CPC, 4 filaments	Evaluation of the Partector 2 Pro for measuring ultrafine particles emitted from 3D printing.	<i>J Occup Environ Hyg</i>	Multi-instrument field evaluation	UFP number concentration and D10-D95 mobility diameter percentiles	Laboratory
Vishakhapatnam long-term co-location	Exploring Environmental and Temporal Performances of Machine Learning Models for Calibration of a Low-Cost PM _{2.5} Sensor.	<i>Aerosol Air Qual Res</i>	Sensor co-location	Low-cost PM _{2.5} against reference, binned by temperature, RH, hour and month	India
Exposure assessment & modelling					

First author	Focus	Journal	Design	Metric	Region
118 CMAQ scenarios, 17 regions x 7 sectors	A LASSO-based reduced-form CMAQ model for predicting ozone and PM _{2.5} responses to emission changes in South Korea.	<i>PLoS One</i>	Chemical transport modelling	CMAQ-simulated surface PM _{2.5} and O ₃ response to sectoral emission perturbation	South Korea
13 Java coal plants, 41,119 daily obs 2015-2025	Predictive modelling of fine-mode combustion aerosol events near Indonesian coal-fired power plants using AE-filtered Himawari-8 satellite data.	<i>Environ Monit Assess</i>	Machine-learning model	Himawari-8 AOD filtered by Angstrom exponent > 1.2 as a fine-mode combustion proxy	Not stated
33 microwave radiometers, China, 2024-2025	Surface-Based Temperature Inversions Climatology Across China from Microwave Radiometer Network and Their Relationship to PM _{2.5} Pollution.	<i>Environ Pollut</i>	Field measurement	PM _{2.5} at hotspot stations against hourly MWR-derived inversion and mixing-layer metrics	China
8 Korean agricultural sites, 547 days	Relationships Between PM _{2.5} and Its Precursor Gases in Agricultural Areas of South Korea Using a Loading-Rate-Based Analysis.	<i>Atmosphere</i>	Field measurement	Hourly PM _{2.5} and NH ₃ , NO ₂ , SO ₂ , expressed as concentration and as wind-scaled loading rate	South Korea
Lahore, 30 filters, Mar-Apr	Carbonaceous Composition and Multi-Wavelength Optical Properties of Particulate Matter in Lahore, Pakistan.	<i>Atmosphere</i>	Measurement campaign	Gravimetric PM _{2.5} , OC/EC, five-wavelength absorption (375-850 nm) by MWAA	Not stated
Melbourne, 3 acellular OP assays, 2 sites	Source Apportionment of PM _{2.5} Oxidative Potential in metropolitan Melbourne: integrating PMF with multiple linear regression across three acellular assays.	<i>Environ Pollut</i>	Source apportionment	PM _{2.5} oxidative potential by DTT, AA and DCFH, regressed on PMF source contributions	Australia
Oradea Metropolitan Area, 4 reference stations	Spatiotemporal assessment of urban PM ₁₀ and population exposure using integrated satellite, meteorological, and socio-spatial data.	<i>Front Public Health</i>	Field campaign / remote sensing	MODIS-MAIAC AOD regressed onto station PM ₁₀ , plus a socio-spatial exposure index	Europe
Shenzhen, 2014-2024 filter series	Unraveling decadal changes in PM _{2.5} sources and secondary aerosol formation drivers using integrated receptor model and machine learning.	<i>J Environ Manage</i>	Source apportionment	Speciated PM _{2.5} filter samples, PMF factors plus ML driver attribution	China
Other clinical endpoints					
Nigeria 3-state cohort, n=5,621	Ambient PM _{2.5} exposure and cancer incidence in Nigerian adults: A prospective cohort study.	<i>Cancer Epidemiol</i>	Prospective cohort	Hybrid satellite + land-use regression, 1 km annual PM _{2.5} , PM ₁₀ , NO ₂	Sub-Saharan Africa
SEER-Medicare wildfire PM _{2.5} , 414,016 lung cancers	Long-term exposure to wildfire-related PM _{2.5} and overall survival of older patients with lung cancer: a retrospective, registry-based cohort study.	<i>Lancet Oncol</i>	Retrospective cohort	3-year moving average wildfire vs non-wildfire PM _{2.5} , zip-code, model-derived	United States

Provenance and method

Entry window **2026-09-15** → **2026-09-15**, contiguous with the 14 September issue; no gap. Built on 16 September under the standing rule that the current day is not opened before 22:00 local. **Sources.** PubMed [Date - Entry], run as two separate queries — health axis and instrumentation axis — through both the local harvester and the connector: **30 unique candidates**, the connector contributing 4 the harvester missed and the harvester 7 the connector missed, with the instrumentation query returning a single record already inside the health set. Europe PMC CREATION_DATE 40, predominantly off-topic MDPI deposits. Crossref windowed by ISSN across the tracked non-PubMed journals: **57 deposits**. **arXiv returned HTTP 406 Not Acceptable** and an empty body — a new failure mode, distinct from the behaviour previously logged for submittedDate. OpenAlex returned HTTP 429 again, as on every recent run. Consensus sweep on low-cost sensor calibration returned 10. **Screening.** 130 raw records, 120 after cross-source dedupe, 111 unseen. **22 shipped, 23 excluded or held** — 7 duplicates, of which **3 were caught by the DOI gate after title screening called them new**; 12 out of scope; and **4 held** whose DOIs resolve but for which no abstract exists in Crossref, Europe PMC, Semantic Scholar, Consensus or Scholar Gateway. The held records are deliberately kept out of `seen.json` so a later issue can carry them. **Trial watch.** One registry record updated in window — NCT07713264 (VISIONS, Stanford University, outdoor workers, interventional, $n = 50$). `analyze_endpoints` was run per protocol: all three registered endpoints are psychosocial scales and **no air-quality or PM measure exists in the protocol**, the second consecutive topical hit whose PM framing is not carried by any measured PM variable. Accumulated to `trials.json` and surfaced in the weekly rollup, not here. **Labels.** Subtopic, design, particle-metric and geography are assigned from title, abstract, keywords and MeSH; assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. One new design label (*Bayesian spatiotemporal model*) was added to the classifier in the same edit that wrote it onto a record; every other label reused the archive. Effect estimates are transcribed verbatim from abstracts and none is rescaled, with the single exception of the Yancheng percent-change, converted to a risk ratio and flagged as such. Content derived from PubMed and Crossref metadata; abstracts remain the copyright of their respective publishers.